

MODULE 24

Data Science Projects and Portfolio

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Data Science Projects and Portfolio

Project 1 Customer Churn Prediction

```
# End-to-end churn prediction pipeline
import pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import classification_report

# Feature groups
num_features = ['tenure', 'monthly_charges', 'total_charges']
cat_features = ['contract', 'payment_method', 'internet_service']

preprocessor = ColumnTransformer([
    ('num', StandardScaler(), num_features),
    ('cat', OneHotEncoder(handle_unknown='ignore'), cat_features)
])
pipeline = Pipeline([('prep', preprocessor), ('clf', GradientBoostingClassifier())])
pipeline.fit(X_train, y_train)
print(classification_report(y_test, pipeline.predict(X_test)))
```

Project 2 House Price Prediction

```
# Feature engineering + regression
import pandas as pd, numpy as np
from xgboost import XGBRegressor
from sklearn.metrics import mean_squared_error

df = pd.read_csv('houses.csv')

# Feature engineering
df['age'] = 2024 - df['year_built']
df['total_sqft'] = df['ground_sqft'] + df['basement_sqft']
df['price_per_sqft'] = df['price'] / df['total_sqft']
df['has_pool'] = (df['pool_area'] > 0).astype(int)

# Log-transform skewed target
df['log_price'] = np.log1p(df['price'])

model = XGBRegressor(n_estimators=500, learning_rate=0.05, max_depth=6)
model.fit(X_train, y_train)
preds = np.expm1(model.predict(X_test))
print('RMSE:', np.sqrt(mean_squared_error(y_test, preds)))
```

Project 3 Image Classifier

```
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import Dense, GlobalAveragePooling2D
```

```

from tensorflow.keras.models import Model
from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Transfer learning
base = MobileNetV2(weights='imagenet', include_top=False, input_shape=(224,224,3))
base.trainable = False

x = base.output
x = GlobalAveragePooling2D()(x)
x = Dense(256, activation='relu')(x)
out = Dense(10, activation='softmax')(x)
model = Model(base.input, out)

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

train_gen = ImageDataGenerator(rescale=1./255, horizontal_flip=True)
model.fit(train_gen.flow_from_directory('data/train', target_size=(224,224)),
epochs=20)

```

Project 4 NLP Sentiment Analysis

```

from transformers import pipeline, AutoTokenizer, AutoModelForSequenceClassification
import torch

# Fine-tune BERT for sentiment
model_name = 'distilbert-base-uncased-finetuned-sst-2-english'
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSequenceClassification.from_pretrained(model_name)

# Inference
sentiment_pipe = pipeline('sentiment-analysis', model=model, tokenizer=tokenizer)
reviews = ['Great product!', 'Terrible service', 'Average quality']
results = sentiment_pipe(reviews)
for review, result in zip(reviews, results):
    print(f'{review} --> {result["label"]} ({result["score"]:.2f})')

```

Building Your Portfolio

Key projects to showcase:

1. End-to-end ML pipeline with deployment
2. NLP project (sentiment, chatbot, summarizer)
3. Computer vision project
4. Data analysis with storytelling
5. Time series forecasting

Portfolio platforms:

GitHub: host code and notebooks
Kaggle: competitions + public notebooks
Tableau/Streamlit: interactive dashboards
Medium/Blog: write about your projects

Resume tips:

Quantify results (improved accuracy by 15%)

Link GitHub and Kaggle profiles

Showcase real-world impact